

Ladder or Satisfaction? Why Gallup and the World Values Survey Disagree on Income and Well-Being*

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Abstract

The two main sources of cross-national data on subjective well-being disagree: life evaluations are much more strongly related to GDP per capita in the Gallup World Poll (Cantril ladder, 0–10) than in the World Values Survey (WVS; life satisfaction, 1–10). In 149 country-years observed by both, log GDP per capita explains 0.49 of the variance of Gallup’s mean ladder but 0.11 of WVS mean satisfaction. Is this due to the question or to the samples? I randomize the wording (ladder vs. satisfaction) and the scale (0–10 vs. 1–10) in an online survey of 11000 respondents in ten high-income countries. The ladder shifts answers by -0.45 points, an effect that varies strongly across countries and fully accounts for Gallup’s lower average level. However, it does not steepen the within-country income gradient, and it explains none of the cross-country variation in the Gallup–WVS gap, which reflects differences in samples, modes and context. Wording effects mostly shift levels; sample differences drive the discrepancy in cross-country patterns.

Keywords: subjective well-being; life satisfaction; Cantril ladder; question wording; survey mode; survey experiment.

JEL codes: I31; C83; C90.

1 Introduction

Cross-country comparisons of subjective well-being rely mostly on two surveys: the Gallup World Poll, which underlies the World Happiness Report, and the World Values Survey (WVS). They measure life evaluation with different questions. Gallup asks respondents to place themselves on a

*Data and code: https://github.com/bixiou/wellbeing_gdp_region [TODO: public repository (e.g. OSF); Gallup data cannot be redistributed]. The survey was approved by the CIRED institutional review board (IRB-CIRED-2025-2) and pre-registered at osf.io/7mzn4. I thank Claudia Senik and participants in the 2024 STATEC Well-being Conference for helpful comments, Armon Rezai and the Vienna University of Economics and Business (WU Wien) for access to the Gallup World Poll data, and Julieta Toffoli for research assistance. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Declaration of interest: none.

ladder whose top represents “the best possible life for you” (Cantril, 1965), on a scale from 0 to 10; the WVS asks “how satisfied are you with your life as a whole these days?”, on a scale from 1 to 10. They also differ in their samples: sampling frames, interview modes, questionnaire context, translations, and the years in which each country is surveyed.

The two surveys paint different pictures of the relationship between income and well-being. Deaton (2008) documented that Gallup’s ladder is strongly related to GDP per capita, more so than WVS measures. The difference is large: in the 149 country-years (83 countries) observed by both surveys the same year, log GDP per capita explains 0.49 of the variance of Gallup’s mean ladder, but only 0.11 of WVS mean satisfaction. Conclusions on whether “money buys happiness” across countries, or on whether a country’s world region predicts well-being better than its income, depend on which survey one uses. [TODO: cite companion paper on region vs. income.]

Which difference is responsible: the question, or the samples? If the question, then the two surveys measure different constructs—for instance, if the ladder elicits thoughts about wealth and status (Nilsson et al., 2024) or evaluative rather than hedonic judgments (Kahneman and Deaton, 2010)—and multiple indicators should be used. If the samples, then the surveys measure the same construct with different errors, and one should favor the dataset with the most reliable and comparable samples. Observational comparisons cannot answer this question, because the question and the sample always differ together (Bjørnskov, 2010).

I answer it with a survey experiment. In an online survey of 11000 respondents in ten high-income countries (Fabre, 2025), I randomly assigned respondents to one of four versions of the life-evaluation question, crossing the wording (Gallup’s ladder vs. WVS life satisfaction) with the scale (0–10 vs. 1–10). The experiment estimates the question effect holding the sample constant. Combining it with Gallup and WVS data for the same countries, I decompose the observed Gallup–WVS gap into a question effect and a residual that captures differences between samples in a broad sense.

The results are clear-cut. The question matters for levels: the ladder shifts answers by -0.45 points on average relative to the satisfaction question, and the effect varies across countries, from -1.12 points in Spain to 0.36 in Japan. This question effect more than accounts for the lower average level of Gallup data in these countries. But it does not account for which countries score relatively higher in Gallup than in the WVS: its covariance with the observed gap is nil or negative, and the residual explains the cross-country variation of the gap. Moreover, the ladder does not steepen the income gradient within countries. Direct comparisons of samples asked the same question confirm that sample and mode effects are as large as the question effect. The experiment covers only high-income countries, so it cannot exclude a larger wording effect in poorer countries, where the gap between the two surveys is largest; but nothing in the evidence points in this direction.

The paper contributes to the methodological literature on the measurement of subjective well-being (Kahneman and Krueger, 2006; OECD, 2013; Stone and Mackie, 2013; Kaiser and Oswald, 2022), in particular on question and scale effects (Conti and Pudney, 2011; Bond and Lang, 2019), context effects (Deaton, 2012; Deaton and Stone, 2016), and mode effects (Dolan and Kavetsos,

2016), and to the debate on the cross-country relation between income and well-being (Deaton, 2008; Stevenson and Wolfers, 2008; Sacks et al., 2012).

2 Data

2.1 World Values Survey

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (Haerpfer et al., 2022), which pools the seven waves conducted between 1981 and 2022: 439531 respondents in 304 country-years and 108 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the United States in 2017 and Japan in 2019). All statistics use the survey weights provided by the WVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: “Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*.”
- *Life satisfaction*: “All things considered, how satisfied are you with your life as a whole these days?”, answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.5 % for satisfaction and 1.7 % for happiness on average) and uncorrelated with GDP per capita (correlation: -0.08); they are excluded.

2.2 Gallup World Poll and World Happiness Report

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (Gallup, 2022). Its life-evaluation question is the Cantril ladder (Cantril, 1965): “Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?”

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2351 country-years, 168 countries, 2006–2023), from which I compute the same satisfaction indicators as for the WVS (mean, shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2). These distributions are unweighted. Gallup’s wave w corresponds to year $w + 2005$: with this mapping, three-year averages of the ladder computed from these distributions correlate at 0.993 with the (weighted) three-year averages published in the World Happiness Report, against 0.89 with the mapping used in earlier versions of this project (year $w + 2000$). To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (World Happiness Report, 2026), which average Gallup data over three years up to 2023–2025.

2.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators ([World Bank, 2026](#)). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are filled in automatically, in three steps: (i) PPP series, which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita; (ii) for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook ([International Monetary Fund, 2026](#)), converted into constant dollars using the U.S. ratio between the World Bank and IMF series; (iii) remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for Northern Ireland). A robustness check excludes all imputed values.

2.4 A survey experiment on question wording and scale

Gallup and WVS differ in two respects that could explain why they disagree: the question (the Cantril ladder on a 0–10 scale vs. life satisfaction on a 1–10 scale) and everything else—sampling frames, interview modes, questionnaire context, translations, and survey years. To separate the two, I embedded a randomized experiment in an original survey ([Fabre, 2025](#)). The survey was conducted online between April 15 and July 3, 2025, by Bilendi (Kantar in Saudi Arabia) on 11000 respondents in 10 high-income countries: the United States (3,000), Japan (2,000), Saudi Arabia (1,000), Germany (1,048), the United Kingdom (826), France (798), Italy (756), Spain (603), Poland (500) and Switzerland (469). (The survey also covered Russia, where the well-being question was not asked.) Samples were stratified with quotas on gender, age, income, education, region and urbanicity (race in the U.S.; citizenship in Saudi Arabia), and results are reweighted to match population frequencies along these dimensions (weights trimmed between 0.25 and 4). Respondents failing an attention test or completing the questionnaire in less than 6 minutes were excluded. The project was approved by the CIRED institutional review board (IRB-CIRED-2025-2) and pre-registered at the Open Science Framework (osf.io/7mzn4).

Toward the end of the questionnaire, after questions on global policies, each respondent was randomly assigned to one of four versions of the life-evaluation question:

- *Ladder 0–10* (Gallup’s original): the Cantril ladder question quoted in Section 2.2, with steps numbered from 0 (*Worst possible*) to 10 (*Best possible*);
- *Ladder 1–10*: same wording, with steps numbered from 1 to 10;
- *Satisfaction 0–10*: the WVS question quoted in Section 2.1, with answers from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*);
- *Satisfaction 1–10* (WVS’s original): same wording, with answers from 1 to 10.

This 2×2 design identifies the effect of wording (ladder vs. satisfaction) and scale (0–10 vs. 1–10), and their interaction, holding the sample, mode, context and time constant. Within each country, the four branches are balanced (each receives between 21% and 30% of respondents).

3 Results

3.1 Experimental effects of wording and scale

Table 1 reports regressions of the answer on the randomized treatments, with country fixed effects, survey weights and heteroskedasticity-robust standard errors. The ladder wording shifts answers by -0.45 points on average (s.e. 0.05), i.e. respondents rate their lives lower on the “best possible life” ladder than when asked how satisfied they are. This is in line with evidence that the two questions elicit different considerations: in open-ended answers, the ladder evokes power and wealth more than life satisfaction does (Nilsson et al., 2024). The scale effect is smaller: answers on the 1–10 scale are 0.14 points higher than on the 0–10 scale (s.e. 0.05), which, once the 1–10 answers are linearly stretched to 0–10, amounts to -0.26 points. Wording and scale do not interact. Measured as the share of respondents answering 6 or more, the ladder changes the share “satisfied” by -9 percentage points.

Table 1: Effects of question wording and scale on life evaluation, Fabre (2025) survey experiment

Dependent Variables: Model:	Answer		Answer (0–10 equiv.)	Satisfied (6+)	Answer (0–10 equiv.)
	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
Ladder wording (Gallup)	-0.451*** (0.046)	-0.455*** (0.067)	-0.456*** (0.067)	-0.088*** (0.013)	-0.464*** (0.123)
Scale 1–10	0.137** (0.046)	0.132 (0.068)	-0.260*** (0.072)	-0.008 (0.013)	-0.276*** (0.046)
Ladder wording (Gallup) $\times$ Scale 1–10		0.008 (0.092)	-0.041 (0.097)	0.028 (0.019)	
Income quartile (1–4)					0.689*** (0.031)
Ladder wording (Gallup) $\times$ Income quartile (1–4)					-0.004 (0.042)
<i>Fixed-effects</i>					
Country	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	11,000	11,000	11,000	11,000	11,000
R ²	0.05681	0.05681	0.05942	0.04208	0.15976

Heteroskedasticity-robust standard-errors in parentheses

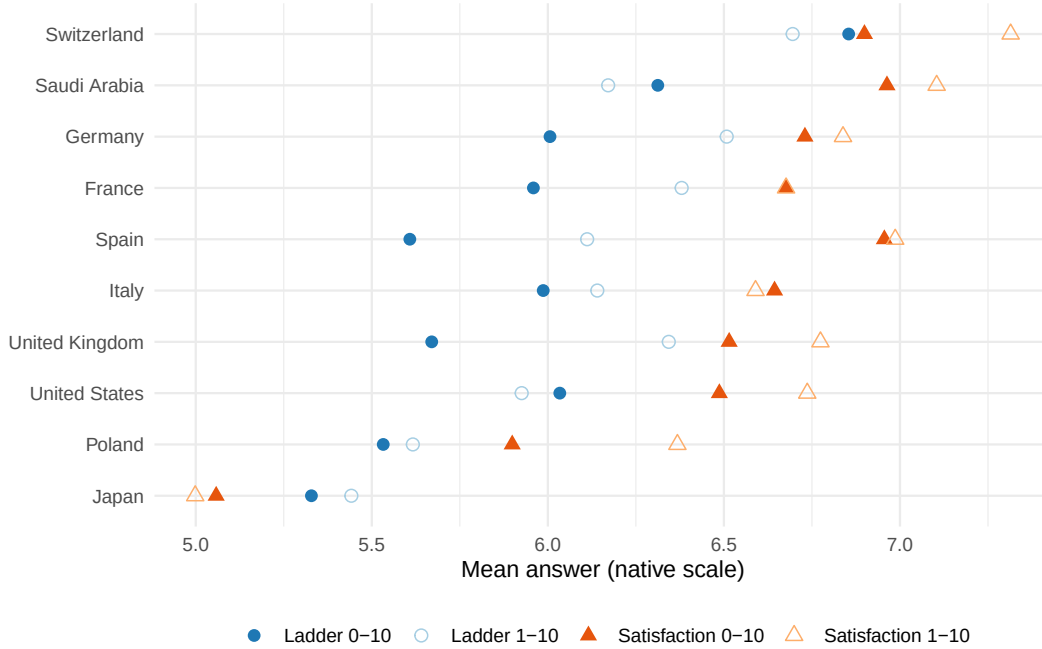
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: OLS with country fixed effects, survey weights (normalized within country) and heteroskedasticity-robust standard errors in parentheses. *Ladder wording:* Cantril ladder (vs. WVS life satisfaction). *Scale 1–10:* answers from 1 to 10 (vs. 0 to 10). In columns 3 and 5, 1–10 answers are linearly rescaled to 0–10. Column 5 interacts the wording with the respondent’s income quartile within their country. * $p < .05$, ** $p < .01$, *** $p < .001$.

Two further results matter for the interpretation of cross-country comparisons. First, the ladder does not steepen the income gradient *within* countries: the interaction between ladder wording and the income quartile is -0.004 ($p = 0.92$), against a gradient of 0.69 points per quartile (column 5).

So, at least among individuals in high-income countries, the ladder is not more “about money” than the satisfaction question. Second, the wording effect differs markedly across countries (Figure 1). It ranges from -1.12 points (Spain) to 0.36 (Japan), where the ladder yields *higher* answers than the satisfaction question, and the hypothesis of a common effect across countries is strongly rejected ($\chi^2(9) = 93.4, p < .001$). Question wording therefore affects not only the level but also the ranking of countries.

Figure 1: Mean answer by question variant and country, Fabre (2025)



Note: Weighted mean answer on the native scale of each variant (0–10 or 1–10).

3.2 Decomposing the Gallup–WVS gap in ten countries

For each country c of the experiment, I define the observed gap between Gallup and the WVS as $D_c = G_c - W_c$, where W_c is the mean satisfaction in the latest WVS survey of the country and G_c the mean ladder in Gallup the same year (or the closest year, at most three years apart; seven countries have an exact match). Comparing the two datasets in the same year removes the influence of changes in well-being over time. The experiment measures, within the same sample, the *question effect* $Q_c = F_c(\text{ladder } 0\text{--}10) - F_c(\text{satisfaction } 1\text{--}10)$, where $F_c(v)$ is the mean answer to variant v in country c . The residual $R_c = D_c - Q_c$ is the part of the gap due to everything that differs between the two datasets except the question: sampling frames, interview modes, questionnaire context, translations, and survey-specific time effects. This decomposition assumes that the question effect measured online in 2025 applies to the Gallup and WVS samples of earlier years.

Table 2 and Figure 2 present the decomposition for each country, and Table 3 summarizes it with 95% confidence intervals from a bootstrap that resamples respondents in the three surveys.

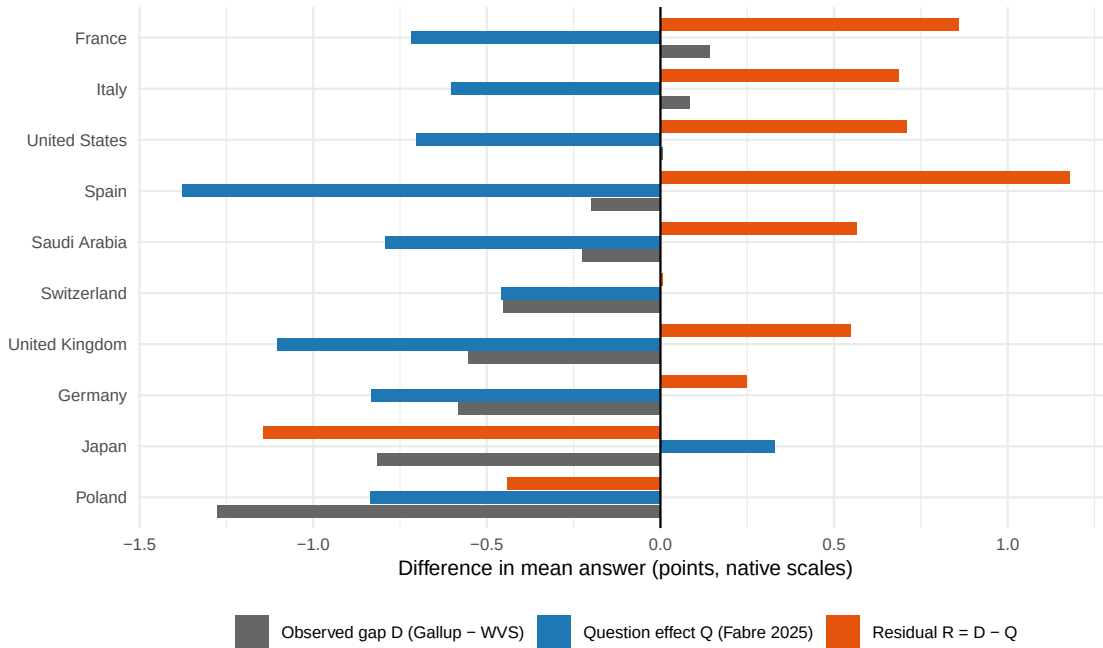
On average, the gap between Gallup and WVS answers is $D = -0.39$ points. The question effect is even larger in absolute value: $Q = -0.71$ points. Wording (and scale) thus fully account for the fact that Gallup yields lower levels than the WVS in high-income countries, and the residual partly offsets it ($R = 0.32$). Since the two components move in opposite directions, another way to summarize the level difference is to compare the size of the two movements: the question accounts for $|\bar{Q}|/(|\bar{Q}| + |\bar{R}|) = 0.69$ (95% CI [0.64; 0.77]) of the gross difference in levels between Gallup and the WVS, and for 0.58 on average when this share is computed country by country. By this metric, wording is the main driver of the difference in *levels*.

Table 2: Decomposition of the Gallup–WVS gap by country

Country	Year	WVS (Gallup)	WVS mode	Gallup (1)	WVS (2)	Gap D (1)–(2)	Ladder 0–10 Fabre (2025)	Satisf. 1–10	Question effect Q	Residual R $D - Q$
Poland	2012		Face-to-face	5.78	7.06	-1.28	5.53	6.37	-0.84	-0.44
Japan	2019		Mail	5.94	6.76	-0.82	5.33	5.00	0.33	-1.15
Germany	2018		CAPI	7.15	7.74	-0.58	6.01	6.84	-0.83	0.25
United Kingdom	2022		CAPI/Web/Mail	6.76	7.32	-0.56	5.67	6.77	-1.10	0.55
Switzerland	2007 (2006)		Face-to-face	7.45	7.91	-0.45	6.85	7.31	-0.46	0.01
Saudi Arabia	2003 (2006)		Face-to-face	7.06	7.28	-0.23	6.31	7.10	-0.79	0.57
Spain	2011		Face-to-face	6.57	6.77	-0.20	5.61	6.99	-1.38	1.18
United States	2017		Web/Phone	7.23	7.22	0.01	6.03	6.74	-0.70	0.71
Italy	2005 (2006)		Face-to-face	6.97	6.89	0.08	5.99	6.59	-0.60	0.69
France	2006		Face-to-face	7.01	6.86	0.14	5.96	6.68	-0.72	0.86

Note: Mean answers on native scales (Gallup: 0–10; WVS: 1–10). The Gallup year is given in parentheses when it differs from the WVS year. WVS mode: main interview mode(s) of the WVS survey (CAPI: computer-assisted personal interviews).

Figure 2: Observed Gallup–WVS gap, question effect and residual by country



What matters for correlations with GDP and for country rankings, however, is not the level of the gap but its variation across countries, which the previous metrics do not capture: a question effect

that shifted all countries by the same amount would score high on them while leaving correlations and rankings unchanged. Here, the question effect explains nothing: the covariance between D_c and Q_c is slightly negative, so that the share of the cross-country variance of the gap attributable to the question, $\text{cov}(D, Q)/\text{var}(D)$, is -0.20 (95% CI [-0.52; 0.15]), and the residual accounts for the rest. Predicting a country's Gallup score by adding its experimentally measured question effect to its WVS score does worse (root mean squared error: 0.73) than simply adding the average gap (0.42). The conclusion is the same when restricting to the 7 countries where Gallup and WVS are observed the same year (variance share: -0.27) or to the 4 countries with a WVS survey since 2017 (-0.81), and for the share of respondents answering 6 or more (-0.11). For instance, the question effect is largest in Spain and the United Kingdom, but the observed gap is small in Spain; Japan is the only country with a positive question effect, but the gap is negative there.

Table 3: Decomposition of the Gallup–WVS gap: summary

Statistic	Estimate [95% bootstrap CI]
Mean gap D (Gallup – WVS)	-0.39 [-0.44; -0.34]
Mean question effect Q (wording + scale)	-0.71 [-0.85; -0.56]
Mean residual R (sampling, mode, context)	0.32 [0.17; 0.46]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.83 [1.42; 2.27]
Share of gross level difference due to question ($ \bar{Q} /(\bar{Q} + \bar{R})$)	0.69 [0.64; 0.77]
Same, country average ($ \bar{Q}_c /(\bar{Q}_c + \bar{R}_c)$)	0.58 [0.48; 0.62]
Share of cross-country variance of D due to Q ($\text{cov}(D, Q)/\text{var}(D)$)	-0.20 [-0.52; 0.15]
Share of cross-country variance of D due to R	1.20 [0.85; 1.52]
Mean absolute gap $ D $	0.43 [0.40; 0.49]
Mean absolute residual $ R $	0.64 [0.53; 0.79]
Correlation between D and Q	-0.20 [-0.47; 0.14]
Mean gap, share satisfied (6+)	-0.05 [-0.07; -0.04]
Mean question effect, share satisfied (6+)	-0.11 [-0.14; -0.08]
Variance share due to Q , share satisfied	-0.11 [-0.44; 0.22]

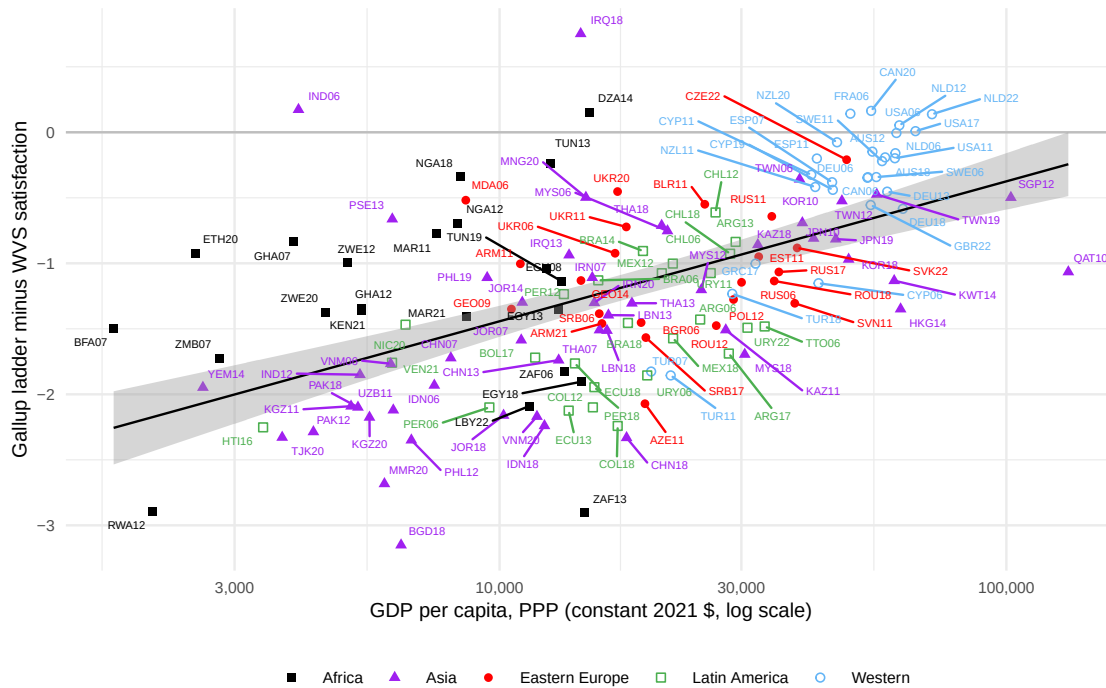
Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country $\times$ variant (Fabre), within country (WVS), and answers within country (Gallup, multinomial).

Two direct comparisons of samples asked the same question corroborate the importance of the residual. First, Gallup respondents (World Happiness Report, 2023–2025) rate their lives 0.75 points higher on the 0–10 ladder than Fabre (2025) online respondents asked the same question, with differences ranging from 0.16 to 1.24 across countries. Second, WVS respondents report 0.54 points higher satisfaction than online respondents asked the WVS question (this comparison also includes changes over time, as WVS surveys are older). These sample (and mode) effects are of the same order of magnitude as the question effect, and they vary across countries. Even in the United States, where the 2017 WVS was itself conducted online, WVS respondents report 0.49 points higher satisfaction than Fabre (2025) respondents.

3.3 The global discrepancy

The ten countries of the experiment are all high-income, whereas the Gallup–WVS discrepancy matters most for the global relation between GDP and well-being. Restricting both datasets to the 149 country-years (83 countries) observed by both surveys the same year, log GDP explains 0.49 of the variance of Gallup’s mean ladder but only 0.11 of WVS mean satisfaction, and income accounts for 51 % of the variance explained jointly with region in Gallup against 15 % in the WVS. The gap between the two datasets increases strongly with GDP (Figure 3): the slope is 1.07 per unit of $\log_{10}$ GDP (s.e. 0.16, clustered by country), with WVS answers exceeding Gallup answers by around two points in low-income countries, against less than half a point in high-income countries.

Figure 3: Gallup ladder minus WVS satisfaction in the same country-year, by GDP per capita



Note: Mean answers on native scales (Gallup: 0–10; WVS: 1–10), 149 country-years observed by both surveys. Line: OLS fit with 95% confidence band.

Could the question explain this pattern? Within the experiment, the question effect is not significantly related to GDP (slope 0.30, s.e. 1.75), which is unsurprising given the narrow range of GDP among the ten countries. Even taking this point estimate at face value and extrapolating it to all countries, adjusting WVS answers for the predicted question effect would raise the R^2 of WVS satisfaction on log GDP only from 0.11 to 0.20, far from Gallup’s 0.49. The experiment cannot rule out that the wording effect is much larger in low-income countries—for instance if the “best possible life” evokes material comparisons more strongly where material deprivation is widespread—but nothing in the high-income evidence points in that direction: the ladder does not steepen the individual income gradient (Section 3.1), and the country-level question effects are unrelated to the observed gaps.

4 Discussion

The experiment shows that the question matters for levels: the Cantril ladder yields lower answers than the WVS life-satisfaction question, and this difference fully accounts for the lower average level of Gallup data in high-income countries. But the cross-country pattern of the Gallup–WVS gap, which is what drives differences in correlations with GDP and in country rankings, is not explained by the question. It is explained by the residual, i.e. by differences between samples in the broad sense: sampling frames, interview modes (Dolan and Kavetsos, 2016; Conti and Pudney, 2011), questionnaire context (Deaton, 2012; Deaton and Stone, 2016), translations, and time.

Several limitations should be kept in mind. First, the experiment covers ten high-income countries, whereas the gap varies most across income levels. The question effect could be different in low- and middle-income countries, where both Gallup and the WVS mostly interview face-to-face; only an experiment in these countries can settle this. Second, the residual lumps together several factors that the design cannot separate. In particular, the question effect was measured with the survey’s own translations, toward the end of a questionnaire on global policies, and online; it may differ from the effect that would be obtained with official Gallup and WVS translations, at the beginning of an interview, and face-to-face or by phone. Third, the decomposition assumes that the question effect measured in 2025 applies to earlier years; for several countries, the latest WVS survey dates from the 2000s. Fourth, with ten countries, cross-country statistics are imprecise, as shown by the bootstrap confidence intervals. Finally, online panel respondents report markedly lower well-being than Gallup and WVS respondents asked the same question, a sample and mode effect that deserves attention in its own right as online panels become common in well-being research.

5 Conclusion

The Gallup World Poll and the World Values Survey disagree on how strongly national well-being is related to income. A survey experiment in ten high-income countries shows that their different questions explain why Gallup’s life evaluations are lower on average, but not why the gap between the two surveys varies across countries. The discrepancy in cross-country patterns therefore comes mainly from differences in samples, interview modes, and context, rather than from the constructs measured.

This has two implications. First, the Cantril ladder and the life-satisfaction question can be treated as measures of the same construct up to a shift in level, at least in high-income countries; the choice between datasets should be based on the quality and comparability of their samples. Gallup’s standardized annual methodology is an argument in its favor for cross-country comparisons, but the two surveys disagree most in low-income countries, where there is not yet evidence on which is closer to the truth. Second, the country-specific wording effects imply that rankings of countries are sensitive to the question asked, even among high-income countries. Replicating the experiment face-to-face in low- and middle-income countries, with official translations and at the beginning of the interview, is the natural next step.

Declaration of generative AI and AI-assisted technologies in the writing process

[TODO: During the preparation of this work the author used Claude (Anthropic) to write analysis code and draft parts of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. (Adapt to actual use.)]

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